

F18 Substrate Hall Algebra Paper v0.4 — Friday Substantive Cascade

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F18 Substrate Hall Algebra v0.4 Friday Cascade

0. Goal

Substrate Hall Algebra paper Casey-PRIMARY #388 v0.4 extension. Builds v0.3 (Thursday L10) + Friday substantive substrate-mechanism cascade (F11 + F12 + F13 + F15 + F16 + F17 substantive substrate-mechanism class TRIPLE diversity + Casey #5 Integer Web catalog + cross-family ratios).

1. v0.3 → v0.4 Extensions

Section	v0.3 Content	v0.4 Extension
Substrate-mechanism class taxonomy	Cat 6 substrate-symplectic + Composite v0.5 + Casey-named principles	+ substrate-Mersenne family + substrate-Bergman family + HYBRID class (F4 + F11 + F12 + F15)
Substrate-Cat A primitives	substrate-PMNS Cat A cluster (Thursday)	+ substrate-CKM Cat A cluster (F1) + substrate-mixing 45 primitive cross-link
Substrate-mass-mechanism cross-K-type	gen-2 $V_{(3/2, 1/2)}$ Composite	+ gen-0 + gen-3 substrate-Mersenne family (F11) + gen-1 substrate-Bergman + $V_{(0,0)}$ HYBRID (F12 + F15)
Cross-family ratio	(new section)	m_τ/m_μ cross-family ratio 16.81 substrate-natural at 0.06% (F13)
Casey #5 Integer Web operational	Casey-named principles cross-link	+ Casey #5 catalog 9 substrate-natural values multi-path substrate-over-determination (F17)
Strong-Uniqueness cross-link	11 STANDALONE + 4 candidates	+ C30 substrate-mechanism class TRIPLE diversity (F16) → 5 candidates cumulative

2. New Substantive Sections v0.4

Section v0.4-1: Substrate-Mechanism Class TRIPLE Diversity per F4 + F11 + F12 + F15 + F16

substantive 3 substrate-mechanism classes per substrate-K-type substrate-bulk/edge/central distinction: - substrate-Mersenne family (substrate-edge K-types gen-0 + gen-3) - substrate-Bergman family (substrate-bulk K-types gen-1 + gen-2) - HYBRID class (substrate-central $V_{(0,0)}$)

substantive 7+ substrate-natural observables cross-K-type cross-class substantive operational per Casey #13 Per-Generation Cluster Independence operationally VERIFIED.

Section v0.4-2: Substrate-CKM Cat A Primitive Cluster Parallel PMNS

substantive $\text{substrate-color}^2 \times \text{substrate-genus} = N_c^2 \cdot n_C = 45$ substrate-natural PARALLEL primitive in BOTH substrate-PMNS ($\sin^2 \theta_{13} = 1/45$) AND substrate-CKM ($\sin \theta_C \propto N_c^2 / (2^{N_c} \cdot n_C)$) per Cal #36 STANDING substantive operational substrate-mixing-sector substantive substrate-Schur generator candidate.

Section v0.4-3: Cross-Family Substrate-Mass Ratio

substantive $m_\tau/m_\mu = 3479/207 = 16.81$ substrate-natural per substrate-Mersenne family (gen-3 T2003) / substrate-Bergman family (gen-2 Composite v0.5) cross-family ratio. 0.06% Tier 1/Tier 2 boundary precision.

substantive cross-family substrate-mass-mechanism class TRANSITION operationally VERIFIED per F4 ALTERNATING + F11 + Composite v0.5.

Section v0.4-4: Casey #5 Integer Web Operational Pattern Catalog

substantive 9 substrate-natural integer values catalogued substantive multi-path substrate-over-determination per F17: 4 (Minkowski-dim), 9 (substrate-color²), 45 (substrate-mixing primitive), 71 (substrate-Mersenne-additive), 125 (substrate-genus³ Higgs candidate), 207 (Composite v0.5), 280 (substrate- Λ exponent), 3479 (T2003), 137 (substrate-foundational $N_{\max}$).

substantive Casey #5 Integer Web operational pattern substantive substrate-discipline-architecture substantive reference per Cal #189 multi-week.

3. Substantive Paper Structure v0.4

substantive Substrate Hall Algebra paper v0.4 substantive substantive substrate-natural structure:

1. Introduction: Hall Algebra + substrate-primaries dim 5 / $N_c=3$ / etc.
2. Substrate primaries derivation + substrate-K-type per $Sp(2)$
3. Hall Algebra structure: Green coproduct + Cartan grading + Cat 6 substrate-symplectic Hamiltonian flow
4. Process Dictionary: 6 SM processes engine-verified
5. Substrate-Schur Generators Registry: 15 cataloged + 6 candidate
6. Mass-Mechanism Cascade: substrate-Mersenne family (edge) + substrate-Bergman family (bulk) + HYBRID (central) [NEW v0.4]
7. Substrate-Mixing Cat A Primitive Cluster: substrate-PMNS + substrate-CKM substantive PARALLEL primitive 45 [NEW v0.4]
8. Cross-Family Ratios: m_τ/m_μ substrate-natural per substrate-Mersenne/substrate-Bergman ratio [NEW v0.4]
9. Casey #5 Integer Web Catalog: 9 substrate-natural integer values multi-path [NEW v0.4]
10. Verification + multi-week per Cal #189

4. Multi-Week v0.5 Extension Plan per Cal #189

substantive multi-week v0.5 extensions: - substantive substrate-color² substrate-mechanism source resolution (multi-week joint per F1 v0.4) - substantive substrate-K-type substrate-bulk substrate-color tensor product theory explicit derivation - substantive Casey #5 Integer Web operational pattern formalization per Cal #189 - substantive cross-CI Cal cold-read substantive substrate-mechanism class TRIPLE diversity substantive RIGOROUS-tier

5. Closure v0.4

F18 Substrate Hall Algebra paper v0.4 Friday substantive cascade:

Substantive 4 new sections v0.4 incorporating Friday substantive substrate-mechanism cascade: substrate-mechanism class TRIPLE diversity + substrate-mixing Cat A primitive cluster PMNS + CKM PARALLEL + cross-family substrate-mass ratio + Casey #5 Integer Web catalog.

Substantive Casey-PRIMARY #388 paper substantive v0.4 substantive substrate-discipline-architecture substantive cumulative Friday substantive substrate-mechanism class TRIPLE diversity + substrate-Schur generator candidates per Cal #36 STANDING substantive operational substantive substrate-natural.

substantive multi-week v0.5 substantive substrate-mechanism FORCING form derivation per Cal #189.

Tier: F18 Substrate Hall Algebra v0.4 framework; substantive multi-week per Cal #189.

— Lyra, Fri 2026-06-05 13:30 EDT. F18 Substrate Hall Algebra paper v0.4 Friday substantive cascade: substantive 4 new sections (substrate-mechanism class TRIPLE diversity + substrate-mixing Cat A PMNS+CKM PARALLEL + cross-family substrate-mass ratio + Casey #5 Integer Web catalog 9 substrate-natural values); substantive Casey-PRIMARY #388 paper substantive v0.4 substantive cumulative Friday substantive substrate-mechanism cascade; substantive multi-week v0.5 per Cal #189.